



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE 321: Theory of Computing

Batch: 18th

Semester: Fall 2022

Date: 04/01/2023

Mid-Term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

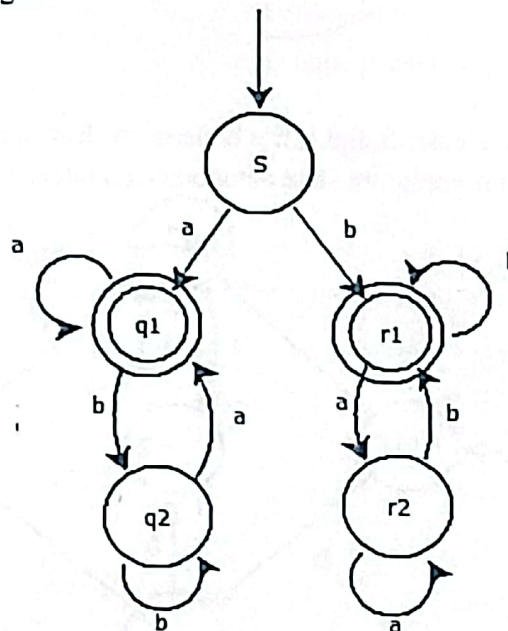
5 x 6 = 30

1. (a) Briefly explain *k-regular* graph with example. 2
- (b) What is Proof by Induction? Prove that, 4

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

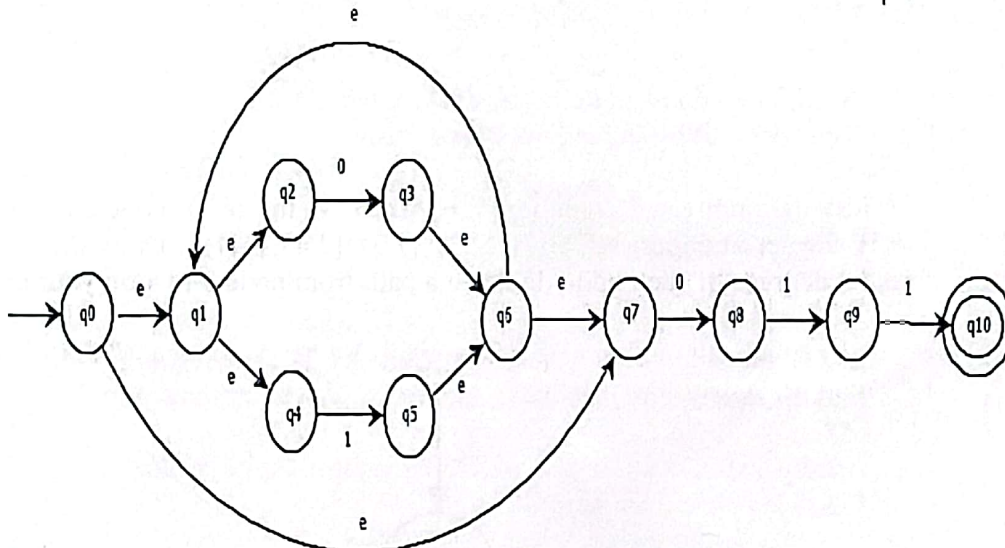
Using Induction Proof.

2. (a) Consider the undirected graph $G=(V,E)$ where V , the set of nodes, is $\{1,2,3,4\}$ 3
and E , the set of edges, is $\{\{1,2\},\{2,3\},\{1,3\},\{2,4\},\{1,4\}\}$. Draw the graph G .
find the degrees of each node. Indicate a path from node 3 to 4 on your drawing of G .
- (b) Briefly explain all tuples of Deterministic Finite Automata (DFA) with the 3
following state diagram-



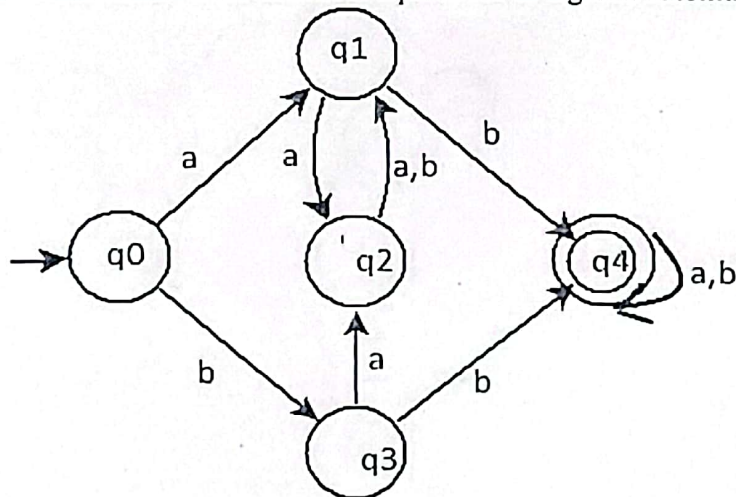
3. (a) Find the differences between Deterministic Finite Automata (DFA) and Non- 2
deterministic Finite Automata (NFA).
- (b) Consider the regular languages, $L(S1) = \{111,1110,10101,11100,100011\}$ and 4
 $L(S2)=\{1011,1101,10110,10111\}$. Draw a DFA state diagram where $L(S1)$ is
accepted by final state and $L(S2)$ is not accepted by final state.

4. (a) What is Kleene closure (Σ^*) and Positive closure (Σ^+)? 2
 (b) Write the regular expression for the following regular languages- 4x1=4
 i. The set of strings over $\{0,1\}$ that have at least one 1.
 ii. The set of strings over $\{0,1\}$ that have at most one 1.
 iii. Regular expression for the set of all strings whose 3rd symbol from the right end is a 0.
 iv. A regular expression for the language of all those strings in which NO 'bab' occurs in the string, defined over $\{a,b\}$.
- 5 (a) Draw a state diagram for every string start and end with alphabet 'a,b' 2
 (b) Convert the following epsilon-NFA to an NFA- 4

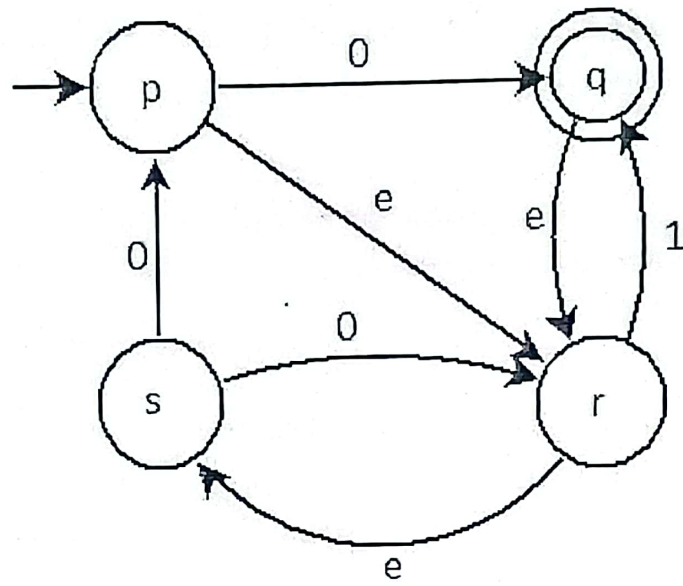


Here, 'e' indicates epsilon sign.

6. (a) If A has a elements and B has b elements, how many elements are in $A \times B$? 2
 (b) Construct a minimum state automata equivalent to a given automata- 4



7. (a) Draw a NFA state diagram for every string ends with '00'. 2
 (b) Convert the epsilon-NFA shown in the following figure to an equivalent DFA. 4



Here, **e** indicates epsilon sign.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 327: Computer Peripherals & Interfacing

Batch: 18th

Semester: Fall 2022

Date: 07/01/2023

Mid Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

- 1 (a) Write the four primary functions that a mouse is capable of. 3
(b) Why did people end up settling on the QWERTY layout for the keyboard? 3
- 2 (a) Define the different types of scanners. 3
(b) What are the basic differences between MICR and OCR? 3
- 3 (a) Outline the advantages of transducer. 3
(b) Explain the working operation of stepper motor. 3
- 4 (a) What is the significance of ASCII code? 2
(b) "A CD-ROM is read-only memory", justify. 2
(c) Write the application of digitized pad and touchscreen. 2
- 5 (a) Describe the scanning techniques used in Raster Scan. 2
(b) Draw the block diagram of a typical computer system. 2
(c) Explain what characteristics a magnetic tape has. 2
- 6 (a) List the devices that provide audio input. 2
(b) Differentiate Raster Scan vs. Random Scan display. 2
(c) Why most computer systems use memory mapped I/O? 2
- 7 (a) Why start and stop bits are needed in asynchronous serial transmission? 2
(b) Write the differences between raster scan and vector scan. 2
(c) How does the GPIB bus function? 2

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 325: System Analysis & Design

Batch: 18th

Semester: Fall 2022

Date: 11/01/2023

Mid-Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Time: 10 Minutes

Part: A

Total Marks: 10 X 1 = 10

[These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.]

1. A professional who use analysis and design techniques to solve business problems involving information technology is a _____.
A) System analysis B) Team leader C) System engineer D) Programmer
E) System Analyst
2. Dividing a system into components based on subsystems which are in turn further divided into subsystems called _____.
A) Interrelated subsystems. B) Information system. C) Depiction of system
D) Functional decomposition. E) System boundary
3. How many types of Information Systems are being used in SAD?
A) Two B) Three C) Four D) Five E) Six
4. About the organization (e.g. company), the system analyst needs not to know _____.
A) What the specific organization does B) What makes it successful
C) Good written and oral presentation D) What its tradition ("culture") and values are
E) What its strategies and plans are
5. Which work should be done at the Planning phases?
A) Analysis B) Information gathering C) Identifying business value
D) Process modeling E) Data modeling
6. How many categories are there in rapid application development?
A) 6 B) 5 C) 4 D) 3 E) 2
7. The organizing and directing of other people to achieve a planned result within a predetermined schedule and budget will be done by the _____.
A) User B) Project Management C) Client D) Project Manager
E) Oversight committee
8. A project manager knows _____.
A) how to plan B) responsibility for the functioning of the team C) execute the plan
D) anticipate problems E) adjust for variances

9. When the Project will launch? A Oversight committee gives _____
 A) Develop a resource plan B) Funds are released and project is announced
 C) Conduct preliminary training and team-building D) Identify and request technical staff
 E) Organize the project team into work groups
10. Normal interviews take lots of time and effort for that a simple technique to define requirements or design a system in a single session by having all the necessary people participating together and is called _____.
 A) Joint Application Design (JAD) B) Joint Design Application (JDA)
 C) Design Application Joint (DAJ) D) Application Design Joint (ADJ)
 E) Application Joint Design (AJD)

Part : B

Time: 1 hour 20 Minutes

Total Marks: 20

(Answer any 04 (Four) questions from the rest)

[The figures in the right margin indicate full marks for the respective question]

- | | | |
|----|---|---|
| 1. | a) Draw the different symbols of DFD | 1 |
| | b) Design the context diagram of course registration system. | 4 |
| 2. | a) Mention the major attributes of the SDLC. | 1 |
| | b) How can we confirm that the project is feasible? | 4 |
| 3. | a) Who are the JAD participants? Briefly explain. | 2 |
| | b) If you want to develop a project how will you make the project schedule? | 3 |
| 4. | a) How do you know "Systems" Thinking? | 2 |
| | b) Define Systems Analysis and Design (SAD)? Why is it important? | 3 |
| 5. | a) As a project manager when you can launch the project? | 2 |
| | b) To solve the problem what are the initiatives will be taken by the analysts? | 3 |

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT323
Mid-Term Examination
Date: 1/14/2023

Course Title: Numerical Analysis
Semester: Fall 2022
Total Marks: 30

Total Time: 1Hours:30 Minutes.

Answer any three (03) of the following questions:

$3 \times 10 = 30$

1. (a) Explain the operator E, ∇ and Δ [3]
(b) Prove the given relation
(i) $\Delta - \nabla = \Delta \nabla$. [3]
(ii) $E \nabla = \nabla E = \nabla$.
(b) Find the approximate root of the equation $f(x) = -\cos x + x$ by using bi-section method. [4]
2. (a) Write down a short note of Newton Raphson's method. [2]
(b) Find the approximate root of the equation $f(x) = x \log_{10} x - 1.2$ by using Newton-Raphson method. [4]
(c) Find the approximate root of the equation $f(x) = x^3 - 4x + 1$ by using false-position method [correct with the five decimals]. [4]
3. (a) State and prove the Lagrange's interpolation formula. [4]
(b) Find the form of the function $y = f(x)$ from the following table [using Lagrange's interpolation formula]: [6]

x	0	1	2	3
$y = f(x)$	-12	0	12	24

4. (a) Derive the Newton's forward formula. [4]
(b) Find the number of death, where the age less than 42. The age and death relation are: [6]
5. (a) Write the condition for a system of equation will be solvable. [4]
(b) Find the solution of the given system of equations: [6]

Age	25-35	35-45	45-55	55-65
Death	13229	18139	24225	31496

$$8x - 3y + 2z = 20.$$

$$4x + 11y - z = 33.$$

$$6x + 3y + 12z = 35.$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 323 : Operating System

Batch: 18th

Semester: Fall 2022

Date: 16/01/2023

Mid-Term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

5 × 6=30

1. (a) Describe the relationship between kernel and shell. 3
(b) Explain the relationships among Hardware, Operating System, Application Software and user with a diagram. 3
2. (a) Define Convey Effect. 1
(b) Instructions Op-codes are as follows: 5
0001 = Load Ac from memory, 0100 = Add Ac with memory, 1000 = Store Ac to memory, 0101 = Multiply Ac by memory, 0010 = Show Ac to Address. Show all the steps of execution of the following Instructions.

Memory	
Address	Data
300	1500
304	4504
308	8504
312	5508
316	2110

PC	
AC	
IR	

Device Address	
110	
111	
112	

Address	Data
500	0022
504	0015
508	0004

3. (a) Explain the execution process of an interrupt with a diagram. 3
(b) How does Operating System (OS) give protection to the processes executing in multiprogramming OS? Explain the processes of verifying the valid instruction for a process. 3
4. (a) Describe the process state information stored in Process Control Block. 2
(b) Consider the 4 processes P_1, P_2, P_3 , and P_4 and their execution times are $C_1=13s$, $C_2=10s$, $C_3=5s$, and $C_4=15s$ respectively. Each of them takes same time for I/O and it is 3s. If the time slice is 3s, then show the execution time line and calculate the total execution time. 4
5. (a) What will happen when a CPU switches its context and put a running process into waiting state for input or output action? Explain the procedure. 3
(b) Short-term scheduler is invoked frequently while long-term scheduler is invoked infrequently. Explain with example. 3

- | | | | |
|----|-----|--|---|
| 6. | (a) | Differentiate between I/O bound process and CPU bound process. | 2 |
| | (b) | Why do we need to maintain the degree of multiprogramming? Explain in details. | 4 |
| 7. | (a) | How does thread increase the execution speed of a process? Explain in details with an example. | 3 |
| | (b) | List different types of threads. Explain them. | 3 |